

10. Consider the following statements:

Statement-I : Rainfall is one of the reasons for weathering of rocks.

Statement-II : Rain water contains carbon dioxide in solution.

Statement-III : Rain water contains atmospheric oxygen.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-II and Statement-III are correct and both of them explain Statement-I
- (b) Both Statement-II and Statement-III are correct, but only one of them explains Statement-I
- (c) Only one of the Statements-II and-III is correct and that explains Statement-I
- (d) Neither Statement-II nor Statement-III is correct

I.A.S. (Pre) 2024

Ans. (a)

Chemical weathering is the process that breaks down rock through chemical changes. Rainfall and temperature can affect the rate at which rocks weather. High temperatures and greater rainfall increase the rate of chemical weathering. Rocks in tropical regions exposed to abundant rainfall and hot temperatures weather much faster than similar rocks residing in cold, dry regions. Rainwater may contain impurities from the atmosphere in the form of dust particles and dissolved gases like oxygen, nitrogen and carbon dioxide.

When water (e.g. rainwater) mixes with carbon dioxide gas in the air or in air pockets in soil, a weak acid solution, called carbonic acid, is produced. When carbonic acid flows through the cracks of some rocks, it chemically reacts with the rock causing some of it to dissolve. This process is called carbonation. It is very active in limestone regions.

Rainwater contains dissolved oxygen from the atmosphere, which can participate in oxidation processes that contribute to weathering. Oxygen in rainwater reacts with minerals, particularly iron-bearing ones, to form oxides or rust, leading to weakening and breakdown.

Hence, both Statement-II and Statement-III are correct and both of them explain Statement-I.

The Volcano

*A volcano is a rupture in the crust of a planetary mass object, such as Earth, that allows hot lava, volcanic ash and gases to escape from a magma chamber below the surface.

*Among the gases emitted through a volcanic explosion the, highest quantity is that of **water vapor** (H₂O). Other gases

emitted are **carbon-dioxide** (CO₂), **sulphur dioxide** (SO₂), **hydrogen** (H₂), **carbon-monoxide** (CO), **hydrogen chloride** (HCL), **hydrogen flouride** (HF) and **helium** (He). **Magma** is composed of molten rock and is stored in the Earth's interior. lava is magma that reaches the Earth's surface with temperature ranging from about 600°C 1200°C. When Lava reaches the surface it cools to form **igneous rocks**. The composition of almost all lava of the earth's crust is dominated by silicate materials. Based on the silicate, lava can be classified into two types- **Acidic Lava** and **Basic Lava**.

***Acidic lava** is highly dense and sticky in nature and has high silica content. Basic lava, on the other hand is less dense and has low silica content. ***Intrusive igneous rocks** are formed when lava cools and solidifies before reaching the earth's surface. Igneous rocks are of two types – intrusive and extrusive.

***Caldera** is an extended version of a volcanic vent. *A **geyser** is a spring characterized by intermittent discharge of water ejected turbulently and accompanied by steam. Generally, all Geyser field sites are located near the active volcanic areas, and the geyser effect is due to the proximity of magma. For example the Old Faithful Geyser of Yellowstone National Park in the United States of America. *A **fumarole** is the last phase in a volcanic activity. It is an opening that emits steam and gases. The name **solfatara** is given to fumarole that emit sulfurous gases. **Mount Katmai** of Alaska (USA) is known as the Valley of Ten Thousand Smokes. Volcanoes are categorized according to their level of activeness–

(1) **Active Volcano** – An active volcano has its vent open and keeps emitting smoke, gas and lava often. At present **Mount Kilauea** is the most active volcano. Situated on Hawaiian Islands of the USA it has been in an active state from between 1983 to 2018.

The top three most active volcanoes are – Kilauea, Mount Etna and Piton de la Fournaise. *On the basis of size, the **largest active volcano** on Earth is **Mauna Loa**, situated on Hawaiian Islands. **Mount Etna** is an active volcano on the east coast of the island of Sicily in Italy. It stands 3357 meters tall. ***Stromboli** is an active volcano on Stromboli Island, north of Sicily in the Tyrrhenian Sea. The volcano has erupted many times and is constantly active with minor eruptions, often visible from many points on the island and from the surrounding sea, giving rise to Island's nickname

“Lighthouse of the Mediterranean.” *Nevados Ojos del Salado is an active volcano in the Andes on the Argentina-Chile border and the highest active volcano in the world at 6879 m. ***Mount Erebus** is an active volcano located on Ross Island in Antarctica. ***Barren Island** is an island located in the Andaman Sea, dominated by Barren Volcano, the only confirmed active volcano in South Asia.

***Mount Fuji** located on Honshu Island in an active volcano in Japan. Presently Mount Vesuvius (Italy) is an active volcano. **Mount St. Helens** is a volcano located in Skamania County, Washington, in the Pacific Northwest region of the United States. It is also known as the '**Fujiyama of America**'.

(2) **Dormant Volcanoes**- A dormant volcano is one that has not erupted recently but is expected to do so again. Kilimanjaro volcano of Tanzania. It has three main volcanic cones- Kibo, Mawenzi and Shira. Among these peaks Kibo is dormant while Mawenzi and Shira are extinct. Damavand of Iran.

(3) **Extinct Volcano** – Extinct volcanoes are those that scientist considers unlikely to erupt again because the volcano no longer has a magma supply. For example- Kulal (Kenya) in Africa, Chimborazo of Ecuador and Popa of Myanmar and Koh-e-Sultan of Pakistan.

1. Consider the following:

- | | |
|-----------------------|----------------------|
| 1. Pyroclastic debris | 2. Ash and dust |
| 3. Nitrogen compounds | 4. Sulphur compounds |

How many of the above are products of volcanic eruptions?

- | | |
|----------------|--------------|
| (a) Only one | (b) Only two |
| (c) Only three | (d) All four |

I.A.S. (Pre) 2024

Ans. (d)

A volcano is a place where gases, ashes and/or molten rock material-lava-escape to the ground. A volcano is called an active volcano if the materials mentioned are being released or have been released in the recent past. The layer below the solid crust is the mantle. The material in the upper mantle portion is called magma. Once it starts moving towards the crust or reaches the surface, it is referred to as lava. The material that reaches the ground includes lava flows, pyroclastic debris, volcanic bombs, ash and dust and gases such as nitrogen compounds, sulphur compounds and minor amounts of chlorine, hydrogen and argon etc. Hence, option (d) is the correct answer.

2. Which of the following is a Basic landform?

- | | |
|-------------------|-------------------------|
| (a) Volcanic cone | (b) Residual mountain |
| (c) Monadnock | (d) Erosional waterfall |

Uttarakhand P.C.S. (Pre) 2016

Ans. (a)

In the given option a Volcanic cone is a basic landform. It is formed as a result of the deposition of lava in a cone around the hole after a volcanic eruption. Significantly residual mountains are that type of mountain in their original form, which remain as residual mountains after being eroded. Similarly, erosional waterfalls are also the product of erosion monadnocks are eroded parts of rocks.

3. The most abundant gas emitted from volcanoes is –

- | | |
|---------------------|--------------------|
| (a) Water Vapour | (b) Helium |
| (c) Sulphur Dioxide | (d) Carbon Dioxide |

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

The most abundant volcanic gas is water vapor (H_2O). Carbon dioxide (CO_2) and sulphur dioxide (SO_2) are two other important gases present in a volcanic eruption. A significant amount of hydrogen sulfide (H_2S), hydrogen (H_2), carbon monoxide (CO), hydrogen chloride (HCl), hydrogen fluoride (HF) and helium (He) gas is also found in it.

4. Which one of the following is the highest volcanic mountain in the world?

- | | |
|--------------------|-----------------------|
| (a) Mount Pinatubo | (b) Mount Kilimanjaro |
| (c) Mount Tall | (d) Mount Cotopaxi |

U.P.P.C.S. (Pre) 2015

Ans. (d)

The highest volcanic mountain of the World in the given options is Mount Cotopaxi with its height of 5911 m. The height of Mount Pinatubo is 1486 m and Mount Kilimanjaro (5881 m). Mount Tall (311 m) is located in Philippines.

5. In which belt, the volcanic eruption is maximum?

- | | |
|---------------------|---------------------------|
| (a) Mid Continental | (b) Circumpacific |
| (c) Mid Atlantic | (d) Intraplates Volcanoes |

M.P. P.C.S. (Pre) 2021

Ans. (b)

Volcanic eruption is highest in the Pacific Ocean belt. About 75 percent of active volcanoes on Earth are located in the Pacific Ocean belt (known as the 'Ring of Fire') and most earthquakes in the world also occur in this region.

6. The molten material found inside the earth is called

- (a) Lava (b) Basalt
(c) Obsidian (d) None of the above

Uttarakhand P.C.S. (Pre) 2006

Ans. (d)

Magma is a molten and semi-molten rock mixture found under the surface of Earth. Magma is a very fluid and dynamic substance having a temperature between about 600°C to 1200°C. When magma is ejected by a volcano, the material is called lava. Magma on cooling converts into a solid known as igneous rock. On the basis of silica's concentration lava is of two types – (1) Acidic lava : It contains a large amount of Silica, highly viscous. (2) Basic lava : It contains a small amount of silica and is highly fluid.

7. Rock formed inside the Earth as a result of solidification of lava is called –

- (a) Plutonic rocks (b) Volcanic rocks
(c) Metamorphic rocks (d) Sedimentary rocks

Uttarakhand P.C.S. (Pre) 2006

Ans. (a)

Igneous rock is formed by cooling and solidification of erupted magma. Internal igneous rocks are basically of two types : (1) Plutonic rock (2) Hypabyssal (intermediary) rock. When magma cools and solidifies below the Earth's surface at great depth the 'Plutonic rocks' are formed. As plutonic rocks are situated deep inside the ground, it takes much time to cool and hence their crystals are of larger size.

8. Volcanic eruptions do not occur in the:

- (a) Baltic Sea (b) Black Sea
(c) Caribbean Sea (d) Caspian Sea

I.A.S. (Pre) 2001

Ans. (a)

The volcanic eruption is related to Plate edges in the convergence and divergence area. In the given option, the Baltic Sea is not related to plate edges, thus, it is clear that volcanic eruptions do not occur in the Baltic Sea.

9. Mt. Etna is –

- (a) A mountain (b) A mountain peak
(c) A volcano (d) A plateau

43rd B.P.S.C. (Pre) 1994

Ans. (c)

Mount Etna is an active volcano on the east-coast of Sicily, Italy. Its maximum height is about 3357 meters.

10. Where is the mount Saint Helens situated ?

- (a) Chile
(b) Japan
(c) Philippines
(d) United State of America

I.A.S. (Pre) 2005

Ans. (d)

Mount Saint Helens is an active strato volcano located in Skamania County, Washington in the Pacific North West region of the United States of America. It is also known as the 'Fujiyama of America'.

11. Mauna Loa is an active volcano in –

- (a) Alaska (b) Hawaii
(c) Italy (d) Japan

U.P.P.C.S. (Mains) 2014

U.P.P.C.S. (Pre) 2005

Ans. (b)

In Hawaiian language, Mauna Loa means the 'Long Mountain'. It is one of the active volcanoes. Mauna Loa had its first well documented eruption in 1843. It is spread over half of the Hawaii Island (USA).

12. Mauna Loa is an example of

- (a) Active volcano
(b) dormant volcano
(c) Dead volcano
(d) Plateau in the volcano region

39th B.P.S.C. (Pre) 1994

Ans. (a)

See the explanation of the above question.

13. Which of the following is the World's largest volcano erupted for the first time in 38 years?

- (a) Haleakala
(b) Mount St. Helens
(c) Mauna Loa
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

The most active volcano in the world, Mauna Loa which is located in Hawaii (US), erupted in 2022, spewing ash and lighting up the sky with a red glow. Significantly the Hawaiian name "Mauna Loa" means "Long Mountain". This name is fitting, because the subaerial (above sea level) part of the volcano extends about 120 km from the southern trip of the island to the summit caldera and then east-northeast to the coastline near Itilo.

14. Which of the following are active volcanoes ?

- | | |
|--------------|-------------|
| 1. Aconcagua | 2. Cotopaxi |
| 3. Etna | 4. Fujiyama |

Select the correct answer from the code given below :

- | | |
|----------------|----------------|
| (a) 1 and 2 | (b) 3 and 4 |
| (c) 1, 2 and 3 | (d) 2, 3 and 4 |

U.P.P.C.S. (Mains) 2007

Ans. (d)

Cotopaxi, Etna and Fujiyama are active volcanoes while Aconcagua is a volcanic mountain but not active volcano. Aconcagua is in Argentina near its border with Chile. Cotopaxi is situated in Ecuador, Etna in Italy and Fujiyama in Japan.

15. Which of the following are active Volcanoes ?

- | | |
|----------------|-------------|
| 1. Aconcagua | 2. Etna |
| 3. Kilimanjaro | 4. Vesuvius |

Select the correct answer from the code given below :

- | | |
|-------------|-------------|
| (a) 1 and 2 | (b) 1 and 3 |
| (c) 2 and 3 | (d) 2 and 4 |

U.P.P.C.S. (Mains) 2011

Ans. (d)

Kilimanjaro is a dormant volcano situated in Tanzania and Aconcagua is a non-volcanic mountain. The other two are active volcanoes. Vesuvius is situated near (Naples) and Etna is in Sicily, Italy. Thus, option (d) 2 and 4 are correct.

16. The highest mountain peak of Africa, Kilimanjaro is situated in –

- | | |
|--------------|------------|
| (a) Kenya | (b) Malawi |
| (c) Tanzania | (d) Zambia |

U.P.P.C.S. (Mains) 2009

Ans. (c)

See the explanation of the above question.

17. Kilimanjaro is a –

- | | |
|-------------|------------|
| (a) Volcano | (b) Island |
| (c) Peak | (d) River |

44th B.P.S.C. (Pre) 2000

Ans. (a)

Kilimanjaro is the highest mountain in Africa. It is situated in the north-eastern part of Tanzania. It has three main volcanic cones- Kibo, Mawenzi and Shira. The highest peak of Mt. Kibo is peak Uhuru whose name was Kaiser-Wilhelm-Spitze before. It is an igneous mountain that forms igneous rocks through erupted lava.

18. Which Volcano is called 'the lighthouse' of Mediterranean Sea ?

- | | |
|---------------|--------------|
| (a) Etna | (b) Pele |
| (c) Stromboli | (d) Vesuvius |

U.P.P.C.S. (Mains) 2004

Ans. (c)

Stromboli is one of the most active volcanoes on the earth located on the island of Stromboli, north of Sicily, southern Italy. Due to the continuous gaseous eruptions, it enlightens the nearby area. Therefore it is known as the 'lighthouse of the Mediterranean.'

19. The Arctic region and the Antarctica continent are situated near:

- | |
|--------------------------------|
| (a) the Amazon Basin |
| (b) the Sahara Desert |
| (c) the North and South Poles |
| (d) More than one of the above |
| (e) None of the above |

68th B.P.S.C. (Pre) 2022

Ans. (c)

The Arctic region and the Antarctica continent are situated near the North and South Poles respectively. The Arctic is the northernmost region on Earth. It is dominated by the Arctic Ocean Basin, and the icy reaches of Scandinavia, Russia, the US of Alaska, Canada and Greenland. Antarctica is a continent. It is Earth's fifth largest continent and is covered completely in ice. Antarctica contains Earth's South Pole.

20. Stromboli is a/an –

- | | |
|---------------------|-----------------------|
| (a) dormant volcano | (b) active volcano |
| (c) extinct volcano | (d) none of the above |

M.P.P.C.S. (Pre) 2017

Ans. (b)

See the explanation of the above question.

21. The most active Volcano is –

- | | |
|--------------|--------------|
| (a) Cotopaxi | (b) Fujiyama |
| (c) Kilauea | (d) Vesuvius |

U.P.P.C.S. (Pre) 2009

U.P.P.C.S. (Mains) 2006

Ans. (c)

The most active volcano in the world is Kilauea on Hawaii Island, United States of America. Kilauea is followed by Etna in Italy and Piton de la Fournaise and Nyamuragira volcanoes on the basis of eruptions .

22. Match List-I with List-II and select the correct answer using the code given below the list:

List-I	List-II
A. Etna	1. Ross Island
B. Vesuvius	2. Ecuador
C. Erebus	3. Italy
D. Cotopaxi	4. Sicily

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	1	2
(c)	3	4	2	1
(d)	4	3	2	1

U.P.P.C.S (Pre) 2010

Ans. (b)

Mount Etna is an active volcano on the east coast of the Mediterranean Sea's Sicily island. Mount Vesuvius is situated at the Bay of Naples, Italy. Mount Erebus is an active volcano on Antarctica's Ross Island. Cotopaxi is located at the Andes mountain peak of Ecuador.

23. Match List-I with List-II and give the correct answer using codes given below –

List – I (Volcano)	List – II (Country)
(A) Mount Rainier	1. Italy
(B) Etna	2. Mexico
(C) Paricutin	3. Philippines
(D) Taal	4. U.S.A.

Code :

	A	B	C	D
(a)	4	2	1	3
(b)	4	1	2	3
(c)	2	1	4	3
(d)	4	3	2	1

I.A.S. (Pre) 1999

Ans. (b)

Mount Rainier is in the USA, Etna is located in Italy. Paricutin is situated in Mexico and Mount Taal is in the Philippines.

24. Match List -I and List-II and select correct answer from the code given below:

List-I	List - II
A. Mount Kinabalu	1. Argentina
B. Elburz	2. Malaysia

C. Aconcagua
D. Kilimanjaro

3. Tanzania
4. Iran

Code:

	A	B	C	D
(a)	1	4	2	3
(b)	3	2	1	4
(c)	2	4	3	1
(d)	2	4	1	3

U.P.P.C.S. (Mains) 2015

Ans. (d)

The correct match is as follows :

Mount Kinabalu	–	Malaysia
Elburz	–	Iran
Aconcagua	–	Argentina
Kilimanjaro	–	Tanzania

25. Match List-I with List-II and select the correct answer from the codes given below the lists:

List - I (Volcano)	List-II (Country)
A. Sabancaya	1. Italy
B. Mount Etna	2. Peru
C. Colima	3. Indonesia
D. Merapi	4. Mexico

Code :

	A	B	C	D
(a)	3	1	4	2
(b)	3	2	1	4
(c)	2	1	4	3
(d)	4	3	2	1

U.P.P.C.S. (Pre) 2015

Ans. (c)

The correct match is as follows :

(Volcano)	(Country)
Sabancaya	Peru
Mount Etna	Italy
Colima	Mexico
Mount Merapi	Indonesia

26. With reference to 'Indonesia', which of the following statements is / are correct?

- (1) It comprises of over 15,000 islands.
- (2) It has over 150 active volcanoes.